Department of Information Technology
Gauhati University

Curriculum for-

1. **B.Tech. (IT/CSE) Honours in Data Science**

Artificial Intelligence and Machine Learning, and Data Science are emerging fields of studies, and are important both from the Students point of view as well as requirements for the Industry and Professions. These disciplines have crossed the boundaries of traditional Information Technology and Computer Science, and becoming Tools and Technologies for across the disciplines. This has created a new demand of specialized training producing empowered human resources with emphasis on the contemporary tools, techniques and applications of AI ML and DS, leaving behind many traditional components of IT/CSE.

While the department shall continue to foster with Core IT/CSE in the B.Tech. curriculum, simultaneously shall facilitate interested local students to be specific in their learning directions with AIML and DS. And for this, these two Honours Programmes are designed and developed.

1. **B.Tech. (IT/CSE) Honours in Data Science**

NEP Based Curriculum 2025

SEMESTER: 1

3 weeks mandatory Induction Programme

SL	Course Code	Course Name	L	T	P	C	Contact Hours	Remarks
1	AS 112	Mathematics-I	3	1	0	4	4	
2	AS 113	Physical Science	3	0	1	4	5	
3	ECE 111	Basic Electrical Electronics Engineering	3	0	1	4	5	
4	IT 111	Fundamentals Of Computing	2	0	1	3	4	
5	ESC 111	Engineering Graphics	1	0	2	3	5	
Total Semester Credit : 18								

Semester: 2

SL	Course Code	Course Name	L	T	P	C	Contact Hours	Remarks
1	AS 121	Chemistry	3	0	1	4	5	
2	AS 123	Mathematics-II	3	1	0	4	4	Including Discrete Mathematics
3	IT 121	Programming For Problem Solving	3	0	1	4	5	
4	BT 121	Modern Biology	2	0	1	3	4	
5	HS 121	English-I	2	0	1	3	4	
6	ESC 121	Workshop	0	0	2	2	4	
7	VAC121	Yoga/NCC/NSS	0	0	2	2	4	
Total Semester Credit : 21								

Semester: 3

SL	Course Code	Course Name	L	T	P	C	Contact Hours	Remarks
1	HS 21X	Foundations on Linguistics/ English -II	2	0	0	2	2	
2	IT 21X	Mathematical and Statistical Foundation for AIML	3	0	0	3	3	
3	ECE 211	Digital Electronics	3	0	1	4	5	
4	IT211	Data Structure and Algorithm	3	0	1	4	5	
5	IT212	IT Workshop	0	0	2	2	4	
6	IT213	Python Programming	2	0	2	4	6	
7	INT 211	Internship	-	-	-	4	-	Carried From 2nd Sem::No Internship Registration For Lateral Entry
Total Semester Credit : 19+4(Internship)								

Semester: 4

SL	Course Code	Course Name	L	T	P	C	Contact Hours	Remarks
1	IT221	Computer Organization And Architecture	3	0	1	4	5	
2	IT225	Python for Data Science	3	0	1	4	5	With OOP
3	IT223	Software Engineering	3	1	0	4	4	
4	IT224	Data Communication & Computer Network	3	0	1	4	5	
5	MG 221	Management-I (Financial And Cost Accounting)	3	0	0	3	3	
6	HS 221	Environmental Sciences	2	0	0	2	2	
	HS22X	Universal Human Values	2	0	0	2	2	
Total Semester Credit : 21+2								

Semester: 5

Sl	Course Code	Course Name	L	T	P	C	Contact Hours	Remarks
1	IT311	Database Management Systems	3	0	1	4	5	
2	IT312	Design And Analysis of Algorithms	3	1	0	4	4	
3	IT313	Operating Systems	3	0	1	4	5	
4	IT315	Foundations of Data Science	3	0	1	4	5	
5	HS 311	Humanities- : Organizational Behaviour	2	0	0	2	3	
6	HS 312	Constitution of India	2	0	0	2	2	
	HS 313	Essence Of Indian Knowledge Tradition						

7	INT 311	DS Based Internship	0	0	0	5		Carried From Sem-4
Total Semester Credit : 20+5(Internship)								

Semester: 6

Sl	Course Code	Course Name	L	T	P	C	Contact Hours	Remarks
1	IT321	Formal Language And Automata Theory	3	0	1	4	5	
2	IT322	Machine Learning	3	0	1	4	5	
3	IT326	Data Analytics	3	0	1	4	3	
4	IT327	Data Visualization and Business Analytics	3	0	1	4	3	
5	HS 321	Foreign Language: French	3	0	0	3	3	
	HS 322	Foreign Language: Russian						
6	IT 323	DS based Project				6		
Total Semester Credit : 25								

Semester: 7

Sl	Course Code	Course Name	L	T	P	C	Contact Hours	Remarks
1	IT411	Deep Learning (Open Elective)	2	0	1	3	4	
2	IT416	Big Data Technologies	3	0	1	4	5	
3	IT417	Data Analytics-2: Data Preprocessing, Filtering and Forecasting	3	0	1	4	5	
4	AS XX	Science Elective				3	3	
5	IT414	ML/DL based DS Project	-	-	-	6	-	
6	IT418	DS Research Methodology	2	0	0	2	-	
7	INT 411	DS Internship	-	-	-	5	-	Carried From Semester 6
Total Semester Credit : 22+5 (Internship)								

Semester: 8

Sl	Course Code	Course Name	L	T	P	C	Contact Hours	Remarks
1	--	Elective	3	0	1	4	4	
2	--	Elective (Open)	2	0	1	3	3	
3	IT421	Project Work/Dissertation				14		
4	IT423	Ethics and Data Privacy	2	0	0	2		
TOTAL SEMESTER CREDIT : 23								

Exit Titles		Entry Criteria	
1st year+ adl course	Certificate in Fundamentals of Science and Technology	1st Year	10+2
2nd year+Adl course	Diploma in CSE/IT	2nd Year	3yrs Diploma
3rd year	B.Sc. (Engineering) in CSE/IT(DS)		
4th year	B.Tech in CSE/IT (Honours in DS)		

List of Electives

1 IT9XX :: Data Mining and Warehousing

2 IT9XX :: Blockchain for Data Science

3 IT9XX :: Financial Data Science and Algorithmic Trading

4 IT9XX :: Climate and Environmental Data Analysis

5 IT9XX::Optimization Techniques for Data Science

6 IT9XX :: Data Governance, Security, and Compliance